LESSON 6.14

The Distributive Property

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use the distributive property to expand and factor expressions.

WHY THIS MATTERS You're selling concert bundles at the music studio. Each bundle includes a ticket ($15) and a snack ($5). If 3 people each buy a bundle, you can calculate the total as 3(15 + 5) OR as 3 × 15 + 3 × 5. Either way, the total is the same — $60! This is the distributive property.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
The Distributive Property *Spanish: La propiedad distributiva*	Multiply a factor by every term inside parentheses.	3(x + 4) = 3x + 12
Distributive Property *Spanish: Propiedad distributiva*	Multiplying a number by everything inside the parentheses: a(b + c) = ab + ac.	3(4 + 5) = 3×4 + 3×5 = 12 + 15 = 27 — the 3 gets 'distributed' to both the 4 and the 5
Factor *Spanish: Factor*	A factor of a whole number divides it evenly, with no remainder. For example, 3 is a factor of 12 because 12 ÷ 3 = 4.	In 3(x + 2), the 3 is the factor outside the parentheses that multiplies each term inside
Expand *Spanish: Desarrollar*	To multiply out the parentheses in an expression.	Expand 2(n + 6): multiply 2 × n = 2n and 2 × 6 = 12, so 2(n + 6) = 2n + 12
Equivalent *Spanish: Equivalente*	Expressions that always have the same value.	3(x + 4) and 3x + 12 always give the same answer: when x = 2, both = 18; when x = 10, both = 42
Coefficient *Spanish: Coeficiente*	The number in front of a letter, like the 3 in 3x.	In 6x + 15, the coefficient of x is 6 — it came from distributing in 3(2x + 5)

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Each bundle has a ticket ($15) and a snack ($5). For 3 bundles I can write 3(15 + 5).

1. Way 1: add inside first. 15 + 5 = 20, then 3 × 20 = 60.
2. Way 2: distribute the 3 to each part. 3 × 15 + 3 × 5.
3. That is 45 + 15 = 60.
4. Both ways give $60, so 3(15 + 5) = 3 × 15 + 3 × 5.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's expand 2(8 + 3). Where does the 2 go?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- When I distribute, I multiply the outside factor _____ by each term _____. The operation inside the parentheses _____ stays the same.

Word bank The Distributive Property | Distributive Property | Factor | Expand | Equivalent | Coefficient | distribute | multiply | each term | factor | expand | equivalent

WATCH OUT!

A common mistake

The most common mistake in The Distributive Property is multiplying the outside factor by only the FIRST term inside the parentheses and forgetting the second

LESSON 6.14

The Distributive Property

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Use the distributive property to expand each expression.

Factored Form	Expanded Form
2(x + 5)	
4(3 + 7)	

HINT Expand means: the outside number multiplies every term inside the parentheses.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Expand 6(n + 3) using the distributive property.

(A) 6n + 18
(B) 6n + 3
(C) 6n + 9
(D) n + 18

HINT The 6 outside must visit BOTH terms inside the parentheses: the n and the 3.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Expand 4(5 − 2) using the distributive property.

(A) 9
(B) 12
(C) 20 − 2
(D) 22

HINT Even with subtraction inside, the 4 must multiply BOTH the 5 and the 2.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Expand 3(x + 4) using the distributive property.

(A) 3x + 12
(B) 3x + 4
(C) 7x
(D) x + 12

HINT The 3 outside multiplies each term inside — both the x and the 4.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

When using the distributive property with a(b + c), you must multiply 'a' by:

(A) Only b
(B) Only c
(C) b + c first, then multiply
(D) Both b and c

HINT This asks about the RULE itself: where does the outside factor 'a' go?

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Expression** 4(n + 3)
2. **Distribute** 4 × n + 3
3. **Simplify** 4n + 3

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Did the 4 get multiplied by both n and 3, or only by the n?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What pattern do you see when you expand an expression using the distributive property?

> **SENTENCE STARTER** When I distribute, I multiply the outside factor ___ by each term ___. The operation inside the parentheses ___ stays the same.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.14

The Distributive Property

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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LEARNING TARGET I can use the distributive property to expand and factor expressions.

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Use the distributive property to expand each expression.

Factored Form	Expanded Form
3(2x + 5)	
7(a − 4)	
2(3m + 8)	
5(6 − y)	

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each factored form to its expanded form.

_____ 2(x + 9)
_____ 5(3n − 2)
_____ 4(a + 6)
_____ 3(7 − y)
_____ 6(2m + 1)
_____ 8(x − 3)

(A) 2x + 18
(B) 15n − 10
(C) 4a + 24
(D) 21 − 3y
(E) 12m + 6
(F) 8x − 24

3 **Write the name of the correct group on each line.**

Expand each expression, then sort by whether the expanded form has a constant greater than 15 or not.

Groups: Constant > 15 | Constant ≤ 15

_____ 3(x + 7)

_____ 2(n + 5)

_____ 5(a + 4)

_____ 4(y − 2)

_____ 6(m + 3)

_____ 2(3x + 1)

4 **Decide whether each pair balances. Check Yes or No, then explain.**

Both equal 32 when x = 5 — they are equivalent by the distributive property.

Left side	Right side	Balanced?

EXPLAIN YOUR CHOICE

PART 2 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Expression** 5(3x + 2)
2. **Distribute** 5 × 3x + 2
3. **Simplify** 15x + 2

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What pattern do you see when you expand an expression using the distributive property?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.14

The Distributive Property

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Work backwards: factor each expanded expression into its factored form using the GCF.

Expanded Form	GCF	Factored Form
6x + 18		
10n − 15		
8a + 12		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Distributive Error

1. **Expression** 5(2x + 4)
2. **Distribute** 5 × 2x + 4
3. **Simplify** 10x + 4

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Use the distributive property to explain why 6 × 98 can be calculated as 6 × 100 − 6 × 2 = 588. Then create your own example of using the distributive property to make multiplication easier.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

What pattern do you see when you expand an expression using the distributive property?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help